Palla Roshan

ic24btech11018@iith.ac.in | chemistryroshan@gmail.com | +91 7416330699 | LinkedIn

EDUCATION

Institution	Degree / Board	Passing Year	Score / CGPA
Indian Institute of Technology Hyderabad	B.Tech	2028	8.38
LAKSHYA SCHOOL SAMALKOTA MANDAL EG DIST AP	Class XII (CBSE)	2024	89.60%
ADITYA (EM) HIGH SCHOOL, GANGARAJU NAGAR EAST GODAVARI DISTRICT	Class X (Board of Secondary Education Andhra Pradesh, India)	2022	95.60%

WORK EXPERIENCE

Research Intern -- Pharmaceutical Solid-State Characterization | *Under Prof. Malla Reddy:*

- Screened 6-10 co-formers through **solvent-mediated co-crystallization** and identified a **novel co-crystal form** of the target API, confirmed via **PXRD** peak-shift and phase-purity analysis; the new form showed a measurable change in stability and dissolution behavior compared to the parent compound.
- Screened API polymorphs across 7-10 solvent systems using **slurry conversion** and **anti-solvent crystallization**, combining **PXRD** peak analysis with **DSC** to distinguish genuine polymorphic transitions from artifacts; controlled crystallization temperature to improve batch-to-batch reproducibility.
- Processed raw PXRD data in **EVA** to extract phase purity and crystallinity values, cross-checking peak assignments against reference databases; selected solvents and cooling conditions based on relevant literature on solid-state pharma chemistry, rather than by trial and error.

PROJECTS

Catalysis & Industrial Chemical Processes | *Under Dr. Natte Kishore and Dr. Sudarsanam Putla:*

- Investigated industrial synthesis of methanol, formaldehyde, and light olefins, analyzing reaction kinetics, catalyst design, conversion, and selectivity across commercial processes.
- Analyzed Haber-Bosch, Fischer-Tropsch, Ziegler-Natta, Friedel-Crafts, and hydrogenation pathways, studying catalyst structure, surface chemistry, active sites, turnover frequency, deactivation, and lab-to-plant scale-up.

Transport Phenomena Analysis | *Under Prof. Ranabir Dey and Prof. Ankush Jaiswal:*

- Applied conservation equations and constitutive relations to momentum, heat, and mass transfer across chemical unit operations, analyzing laminar/turbulent flow, heat exchange, and diffusion.
- Used Reynolds, Nusselt, Prandtl, and Sherwood numbers to characterize transport behavior and evaluate implications for reactor performance, separation efficiency, and equipment design.

Conductive Polymers & Coordination Chemistry | *Under Dr. Priyadarshi Chakraborty:*

- Studied doping mechanisms and soliton-based charge transport in polyacetylene, polypyrrole, and polyaniline, analyzing how doping level and polymer backbone structure influence electrical conductivity.
- Applied crystal field stabilization energy (CFSE) to relate ligand-field effects to transition-metal complex geometry and electronic structure; reviewed conductive polymer applications in flexible electronics and biomedical sensors.

SKILLS

Characterization Techniques: PXRD (polymorph and co-crystal identification), DSC, SEM, TEM, STEM.

Software: EVA for PXRD peak and phase analysis, **ChemDraw** for chemical structure drawing, **GaussView** for molecular modeling and visualization, MS Excel for data logging and event logistics tracking, **PowerPoint** and **Canva** for technical presentation design across 5+ courses and institute events.

RELEVANT COURSES

Nanomaterials & Conductive Polymers, Reaction Engineering, Kinetics & Phase Equilibria, Transport Phenomena & Thermodynamics, Heterogeneous Catalysis, Industrial Chemistry Processes & Coordination Chemistry, Numerical & Computational Chemistry, and AI in Chemistry, Petrochemicals, Petroleum Refining & Plant Design.

POSITIONS OF RESPONSIBILITY

IISM Hospitality Coordinator: Coordinated accommodation and logistics ensuring food and beverages for 2,500+ student athletes and 2,100+ officials across multiple events, working closely with the transport, events and operations under tight scheduling constraints.

EBSB Multimedia Head: Produced photography, videography, and social media content for EBSB events by coordinating well with the team , creating visual and promotional material that documented event activities and drove audience engagement across campus.

Diesta PR Head: Led a social media PR campaign of 20+ posts and reels averaging 7,000+ views each. Shot personal photography for event coverage when needed and made sure that the events got quite decent participation.

OCS Internship Coordinator: Coordinated with recruiters on campus visits and internship drives. Reviewed resumes for 7 students against specific role requirements. Networked with a lot of people from the industry in linkedin.

Milan Hospitality Head: Will plan guest/official accommodation and serve as hospitality point-of-contact; oversee venue setup (competition, practice, spectator areas), ensure safety compliance & bookings; procure and maintain event equipment and supplies.

EXTRACURRICULAR

Photography: Covered institute fests (Diesta, Milan) and EBSB events; personal work in landscape and sky photography.

Music: Extensive music listener and my reference is pop music across English, Telugu, and Tamil.

Sunshine Mentor: Delivered one-on-one peer mentorship and well-being support to 15-30 mentees after completing formal training by being with them throughout the year and helping them out whenever they needed any academic advices.